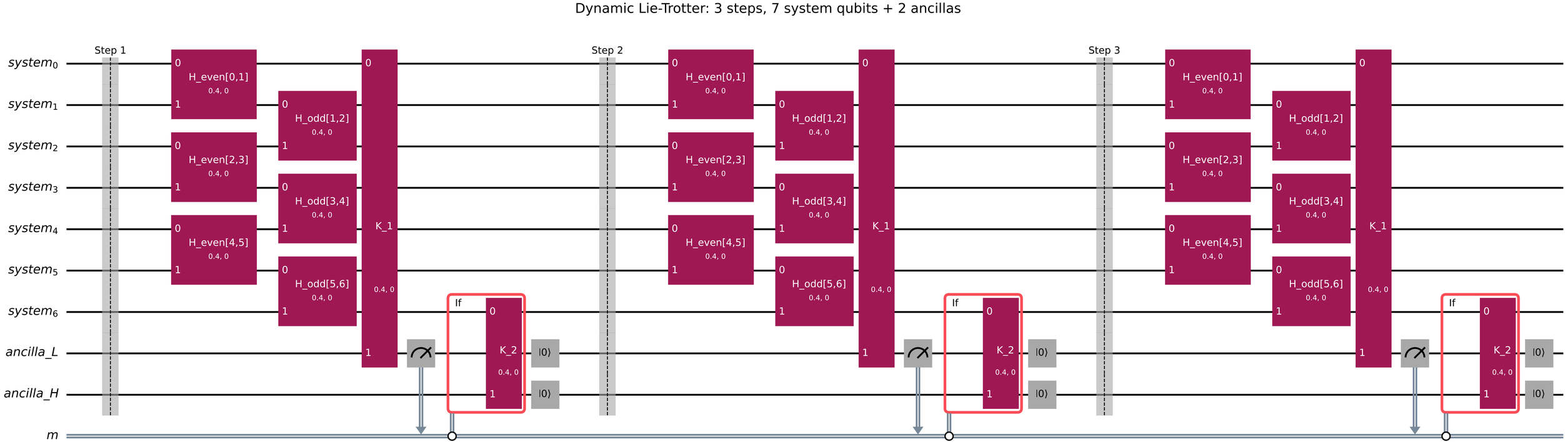
Dynamic Lie-Trotter: 3 steps, 7 system qubits + 2 ancillas
Step 1
Step 2
Step 3
system0
system1
system2
system3
system4
system5
system6
ancilla_L
ancilla_H
m
H_even[0,1]
H_even[2,3]
H_even[4,5]
H_odd[1,2]
H_odd[3,4]
H_odd[5,6]
0.4, 0
K_1
K_2
If
|0⟩
system[i] = model site i = original q[6-i]; H_even/odd[i,j] acts on system[i], system[j].
ancilla_L = low encoding qubit; ancilla_H = high encoding qubit; m is reused; K_2 acts only if m=0. dt=0.2, J=1, gamma=0.2.